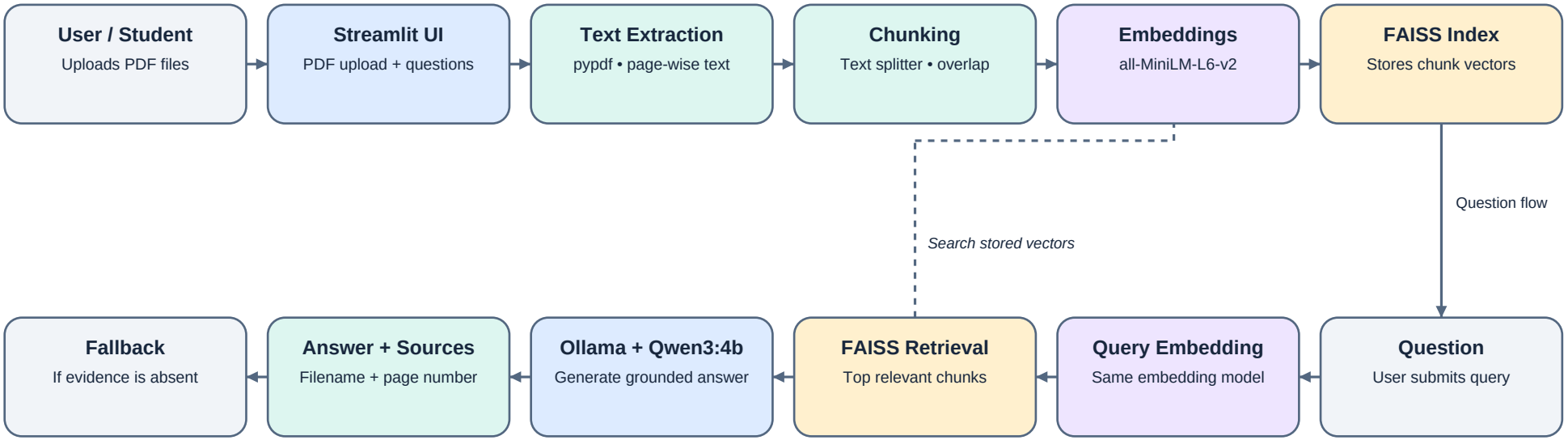
DOMAIN-SPECIFIC RAG CHATBOT

Architecture Diagram • PDF Question Answering • Local deployment



Key behaviour

1. PDFs are parsed page by page; chunks retain document and page metadata.
2. Sentence Transformers creates embeddings; FAISS finds the most relevant chunks for each question.
3. Ollama runs Qwen3:4b locally and generates an answer using retrieved context.
4. The interface shows source filename/page and returns a fallback when the uploaded documents do not contain the answer.

Current local architecture; cloud deployment is not included.